

Roll No.....

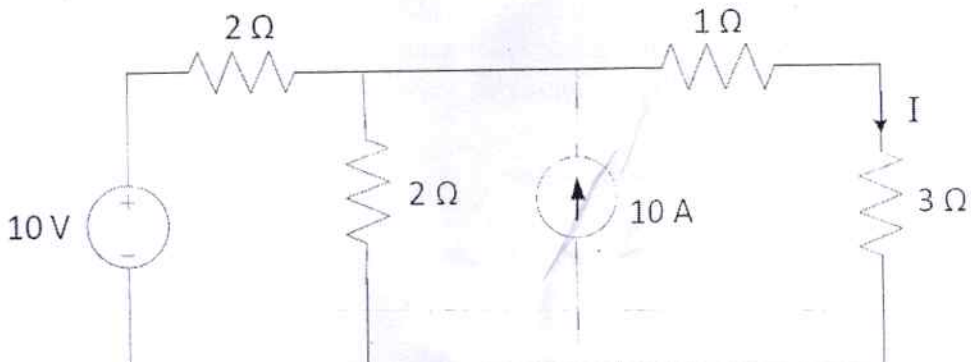
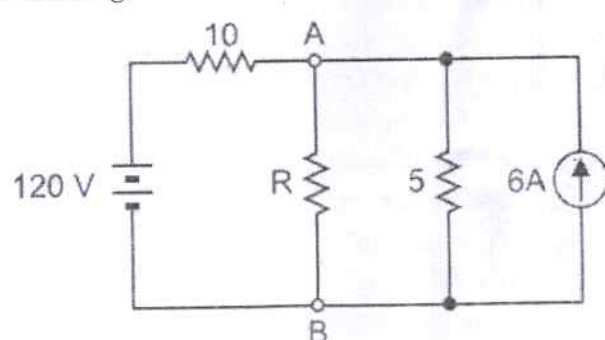
Odd End Semester Examination-2025
B.Tech (Boitech) 1 Semester
Basic Electrical and Electronics Engineering- EEC-101

Time: 3.00 Hrs

Maximum Marks: 100

Note:

- (i) Answer all the questions by choosing any two of the sub questions.
- (ii) Each question carries 10 marks

Q 1	(10 marks Each)	
(a)	Use source Transformation technique to find the load current "I" in the circuit shown below:  Fig. 1	
(b)	(i) Define with suitable examples :Non-linear circuit, Passive element, Mesh, Junction, Superposition theorem (ii) Derive the condition for maximum power transfer in a DC circuit and explain what happens to the efficiency under this condition. <i>Find the value of resistance R to have maximum power transfer in the circuit shown in Fig.</i> 	CO1
(c)	Determine the voltage at point a and 2 by using Nodel analysis	

Q 2	(10 marks Each)	
(a)	(a) Derive an expression for RMS values of sine wave form. (b) An alternating current is expressed as $i = 14.14 \sin 314t$. Determine. i. Maximum current ii. rms current iii. Frequency iv. Instantaneous current when $t = 0.02\text{msec}$.	
(b)	Explain resonance for series RLC circuit and derive the equation for resonant frequency. (b) A series RLC circuit of $R=50\text{ ohms}$, $L=j25\text{ ohms}$. Determine the value of capacitive reactance and impedance at resonance	
(c)	Find Thevenin's equivalent circuit to the left of terminals AB in Fig	CO2
Q 3	(10 marks Each)	CO3
(a)	Explain the concept of a magnetic circuit. Derive the relationship between magnetomotive force (MMF), magnetic flux (Φ), and reluctance ($\mathcal{R}$). Compare it with an electric circuit using suitable analogies and examples.	
(b)	Describe the principle of operation of a single-phase transformer. Draw and explain the phasor diagram under no-load condition, clearly indicating the phase relationship between voltage, flux, and current components.	
(c)	With neat phasor diagrams, explain the behavior of purely resistive, purely inductive, and purely capacitive A.C. circuits. Derive the expressions for voltage, current, and instantaneous power in each case, and show that only the resistive circuit consumes real power.	
Q 4	(10 marks Each)	CO4
(a)	What is semiconductor. Explain about formation of n type and p type semiconductors. Explain the formation of depletion region in pn junction.	

(b)	Explain the working of a half wave rectifier. Derive an expression for RMS current, Average current, efficiency, and ripple factor.	CO5/CO 6
(c)	Explain the working of a center tapped full wave rectifier. Derive an expression for RMS current, Average current, efficiency, and ripple factor.	
Q 5	(10 marks Each)	
(a)	What are filters? Explain the working of capacitor filter, inductor filter and CLC filter with suitable diagram and waveforms.	
(b)	Explain the working of common base configuration of BJT along with its input and output characteristics.	
(c)	Explain the working of common emitter configuration of BJT along with its input and output characteristics.	

